

PHL1010: Practice Problems II
(03/11/2025)

1. A circular loop of radius a carrying current I lies in the x - y plane with its center at the origin.
(a) Write down the expression of $\vec{J}(\vec{r})$ in terms of Dirac-delta function. (b) Find the magnetic field at a point on the z -axis. Use the general form of Biot-Savart Law:

$$\vec{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r'^2} d\tau'$$

- (c) Use the result obtained in (b) to find magnetic field at point P (Figure 1) on the axis of a tightly wound solenoid having n turns per unit length wrapped around a cylindrical tube of radius a and carrying current I . How will your result change if the solenoid is infinitely large?

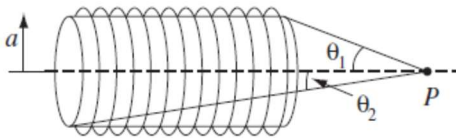


Figure 1

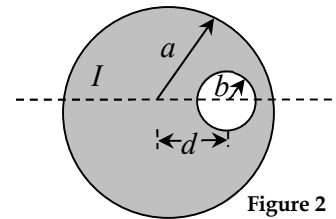


Figure 2

2. Find the magnetic dipole moment of a spherical shell of radius R , carrying a uniform surface charge σ , which is spinning at an angular velocity ω . Show that for points $r > R$ the potential is that of a perfect dipole.
3. A solid sphere of radius R contains a total charge Q distributed uniformly throughout its volume. Sphere is centered at the origin and spins at an angular velocity ω . Assume that the charge distribution is not altered by the rotation. Find (a) the magnetic dipole moment of the sphere, (b) vector potential $\vec{A}$ and (c) the magnetic field inside the sphere (use the findings of Example 5.11 of Griffiths).
4. A long, straight, cylindrical conductor of radius a has a hole of radius b along its length. The axis of hole is shifted by a distance d from the axis of the conductor (see Figure 2). The conductor carries a uniform current I which flows out of the paper. Calculate magnetic field inside the hole. {Hint: use principle of superposition}.
5. A thin glass rod of radius R and length L carries a uniform surface charge σ . It is set spinning about its axis at an angular velocity ω . Find the magnetic field at a distance $s \gg R$ from the center of the rod (Fig. 3). [Hint: treat it as a stack of magnetic dipoles.]

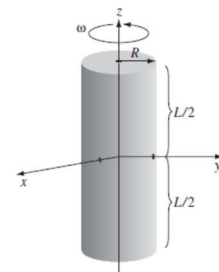


Figure 3